

Lab 01: linear algebra + computing review

Ikea furniture

Your name + Your teammates' names

2026-08-31

Setup

Load packages and data:

```
library(tidyverse)
library(tidymodels)
ikea <- read_csv("data/ikea.csv")
```

Linear algebra review

knitr::include_graphics("exercise1.jpg")

1. a: 3×2 2. a) is ii, This is the only combination where row ^{dimensions} match with column.

b: 2×4

c: 2×3 b) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}$

d: $m \times p$

e: $n \times n$ $\begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 6 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 7 & 11 & 15 \\ 5 & 11 & 17 & 25 \end{bmatrix}$

3. a: True, same dimensions, add all corresponding elements together.

b: True, This is the distributive law of multiplication.

c: False. This is a weird form of distributive law. ~~Matrix~~ Matrix multiplication is irreversible.

4. a) X is non-singular. The dimension is not $n \times n$ but $n \times p$.

b) We don't have enough condition to determine if $X^T X$ is singular. This is because ~~if~~ we don't know if the row or column is depend linearly dependent or not.

5. a) $a-b = \begin{bmatrix} a_1 - b_1 \\ \vdots \\ a_n - b_n \end{bmatrix}$ b. a) $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $X_3 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

$(a-b)^T(a-b) = \sum_{i=1}^n (a_i - b_i)^2$ We want: $a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$ however we can't find corresponding a_1, a_2, a_3

$\therefore$ This is a full column rank matrix.

b) $X_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $X_3 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

We want: $a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$. We can't find corresponding a_1, a_2, a_3 .

$\therefore$ This is a full column rank matrix.

c) $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $X_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $X_4 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

We want: $a_1 X_1 + a_2 X_2 + a_3 X_3 + a_4 X_4 = 0$.

Let $a_1 = 1, a_2 = -1, a_3 = -1, a_4 = 1$:

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 0$

$\therefore$ This is not a full column rank matrix.

The answer 1e changed into P^*P .

Computing review

Exercise 7

```
# add code here
```

...

Exercise 8

...

Exercise 9

...

Exercise 10

...